

BUREAU OF INDIAN STANDARDS

INDIAN STANDARD: IS 1786 : 2008

HIGH STRENGTH DEFORMED STEEL BARS AND WIRES FOR CONCRETE REINFORCEMENT — SPECIFICATION

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1 SCOPE

Covers requirements of high strength deformed steel bars and wires for use as reinforcement in concrete structures.

Table 1: Chemical Composition (Clause 4.2)

Sl No.	Constituent	Fe 500 Max %	Fe 500D Max %
1	Carbon (C)	0.30%	0.25%
2	Sulphur (S)	0.055%	0.040%
3	Phosphorus (P)	0.055%	0.040%
4	Sulphur + Phosphorus (S+P)	0.105%	0.075%

Table 3: Mechanical Properties (Clause 8.1)

Sl No.	Property	Grade Fe 500	Grade Fe 500D
1	0.2% Proof Stress / Yield Strength	Min 500 N/mm ²	Min 500 N/mm ²
2	Tensile Strength (TS) / Yield Strength (YS)	Min 1.08	Min 1.10
3	Elongation (Gauge Length 5.65√A)	Min 14.5%	Min 16.0%